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Studierichting: Informatica
Oefeningenreeks 5L nr. 1.2

$$f(x) = \frac{1}{3}x^3 \text{ op het interval } [0, 1]$$

$$f'(x) = x^2$$

$$(f'(x))^2 = x^4$$

$$S = 2\pi \int_0^1 \frac{1}{3}x^3 \sqrt{1+x^4} dx$$

$$= \frac{2\pi}{3} \int_0^1 x^3 \sqrt{1+x^4} dx$$

$$u = 1 + x^4 \Rightarrow du = 4x^3 dx \Rightarrow dx = \frac{du}{4x^3}$$

$$= \frac{2\pi}{12} \int_1^2 u^{\frac{1}{2}} du$$

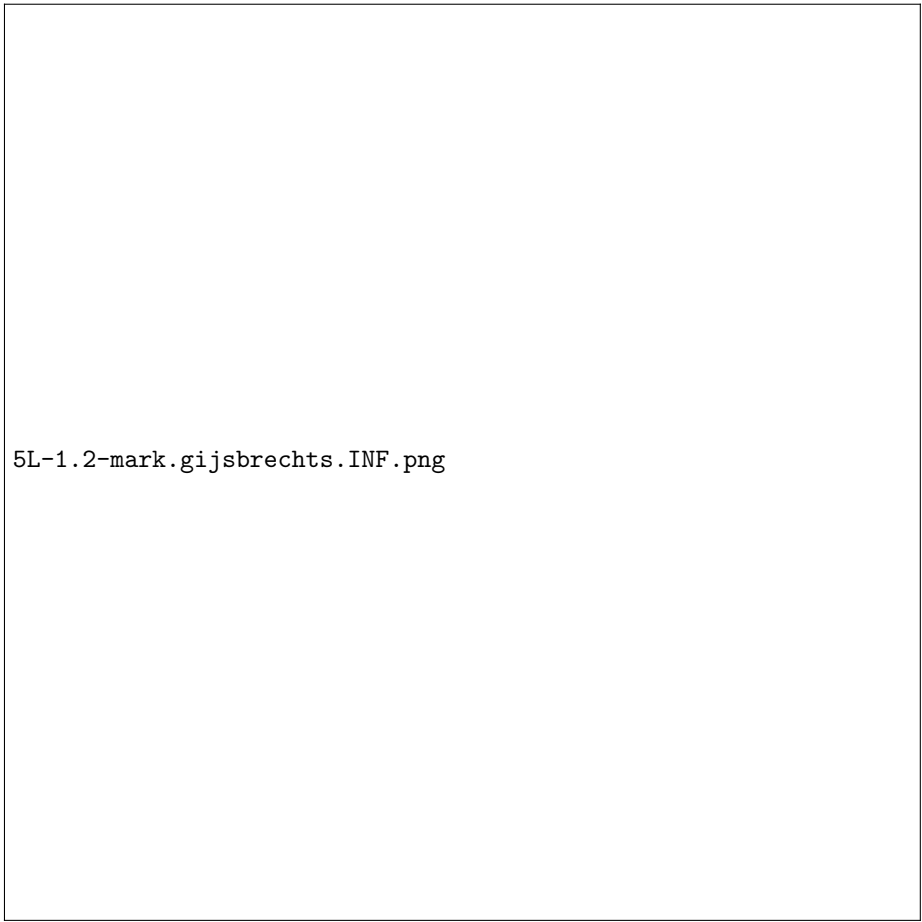
$$= \frac{\pi}{6} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_1^2$$

$$= \frac{2\pi}{18} \left[u^{\frac{3}{2}} \right]_1^2$$

$$= \frac{2\pi}{18} [\sqrt{8} - 1]$$

$$= \frac{\pi}{9} (2\sqrt{2} - 1)$$

References



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